SECOND-ORDER DE BRANGES–NEVANLINNA KERNELS, DERIVATIVE-WRONSKIAN SPLITTING, DAVENPORT–HEILBRONN BLIND SPOTS, AND THE DIFFERENTIAL THRESHOLD β^*

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Abstract. For a real-entire function $F(z)$ and couplings $\alpha, \beta > 0$, we study the second-order structure function $E_{\alpha,\beta}(z) := F(z) - \mathrm{i}\alpha F'(z) - \beta F''(z)$ and its de Branges–Nevanlinna kernel $K_{\alpha,\beta}(z_1, z_2)$. On the diagonal $w = t - \mathrm{i}d$, all three squared moduli and $-2\beta\mathrm{Re}(\overline{F}F'')$ cancel identically, giving $K_{\alpha,\beta}(w,w) \equiv \frac{\alpha}{\pi}\widetilde{W}_{0,1}(d,t) + \frac{\alpha\beta}{\pi}\widetilde{W}_{1,2}(d,t)$. For Riemann's Ξ-function, this lifts the $d = 0$ Lehmer near-coalescence degeneracy at order d^0, couples $\widetilde{W}_{1,2}$ to $\widetilde{W}_{0,1}$ via a radial four-dimensional Laplace operator, and extends to order-M Hurwitz chains. Crucially, however, Hermite–Biehler positivity of $E_{\alpha,\beta}$ for fixed $\beta > 0$ corresponds only to real zeros of $F - \beta F''$, strictly weaker than real zeros of F. Evaluating $K^{\mathrm{DH}}_{\alpha,\beta}$ on the Davenport–Heilbronn function exposes an explicit blind spot: at its critical-line defect $\tau_{\mathrm{false}} \approx 85.7472$, $\pi K^{\mathrm{DH}}_{1,\beta} > 0$ for all $\beta > 0.053495$, motivating the open differential threshold $\beta^*(F) := \inf\{\beta \in \mathbb{R} : F - \beta F'' \text{ has only real zeros}\}$.

1. Introduction and main results

In the analytic theory of Hilbert spaces of entire functions [6, 7, 8] associated with Riemann's completed zeta function $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$ ($z = t - \mathrm{i}d$, $s = \frac{1}{2} + d + \mathrm{i}t$) [22, 20, 26], linear and higher-order differential combinations of $\Xi(z)$ provide natural test families for Hermite–Biehler positivity [14, 16, 17]. While Conrey and Li [2] proved that global inner-product positivity conditions on classical Sonine–de Branges spaces fail for $\xi(s)$, Lagarias [16, 17] studied the first-order structure function $E_\alpha(z) := \Xi(z) - \mathrm{i}\alpha\,\Xi'(z)$ ($\alpha > 0$), whose diagonal de Branges–Nevanlinna kernel reduces to the parity-reduced transverse Wronskian

$$K_\alpha(t - \mathrm{i}d,\, t - \mathrm{i}d) := \frac{|E_\alpha(t - \mathrm{i}d)|^2 - |E^{\#}_\alpha(t - \mathrm{i}d)|^2}{4\pi d} \equiv \frac{\alpha}{\pi}\,\widetilde{W}_{0,1}(d,t), \tag{1.1}$$

where $W_{0,1}(d,t) := \frac{1}{2}\partial_d|\Xi(t - \mathrm{i}d)|^2$ and $\widetilde{W}_{0,1}(d,t) := d^{-1}W_{0,1}(d,t)$ [25, 12, 3, 11]. By Hinkkanen's monotonicity criterion [12] and Lagarias's positivity theorem [16], E_α belongs to the Hermite–Biehler class with respect to $\mathbb{C}_- = \{t - \mathrm{i}d : d > 0\}$ if and only if the Riemann Hypothesis holds.

On the critical line $d = 0$ itself, however, the first-order kernel (1.1) exhibits a local near-coalescence degeneracy whenever two simple zeros approach a Lehmer pair [18]: at a hypothetical double zero $t_0 \in \mathbb{R}$ ($\Xi(t_0) = \Xi'(t_0) = 0$, $\Xi''(t_0) \neq 0$), the critical-line Laguerre invariant $\pi K_\alpha(t_0, t_0) = \alpha((\Xi'(t_0))^2 - \Xi(t_0)\Xi''(t_0)) = \alpha c_1(t_0) = 0$ vanishes at order d^0, so positivity appears only transversely at order d^2 via $c_3(t_0) = \frac{1}{2}(\Xi''(t_0))^2 > 0$. Equivalently, at the local extremum t_{mid} ($\Xi'(t_{\mathrm{mid}}) = 0$) between a closely spaced Lehmer pair $t_\pm = t_{\mathrm{mid}} \pm \delta$ ($\delta \to 0$), $\pi K_\alpha(t_{\mathrm{mid}}, t_{\mathrm{mid}}) = -\alpha\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) = \mathcal{O}(\delta^2) \to 0$.

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A natural question is what happens both algebraically and zero-theoretically when one passes from $E_\alpha(z)$ to the second-order differential structure function

$$E_{\alpha,\beta}(z) := F(z) - \mathrm{i}\alpha\, F'(z) - \beta\, F''(z) \qquad (\alpha > 0,\ \beta > 0), \tag{1.2}$$

including the repeated-root operator $(1 - \frac{\mathrm{i}\alpha}{2}\partial_z)^2 F(z)$ ($\beta = \alpha^2/4$). In this paper we give a complete two-sided answer. On the algebraic side, we prove that $K_{\alpha,\beta}$ splits without cross-term contamination into $\frac{\alpha}{\pi}\widetilde{W}_{0,1} + \frac{\alpha\beta}{\pi}\widetilde{W}_{1,2}$, lifts the $d = 0$ Lehmer near-coalescence degeneracy at order d^0, and satisfies a radial four-dimensional Laplace PDE and dipole recursion. On the zero-theoretic side, we prove that *unlike* $E_\alpha(z)$, Hermite–Biehler membership of $E_{\alpha,\beta}(z)$ for fixed $\beta > 0$ does *not* imply that $F(z)$ has only real zeros: it corresponds only to real zeros of $A_\beta(z) := F(z) - \beta F''(z)$ interlacing with $F'(z)$ [6, 14, 21]. Evaluating $K_{\alpha,\beta}$ on the self-dual Davenport–Heilbronn function [5] demonstrates that for $\beta > 0.053495$ (including $\beta = \frac{1}{4}$), the positive second-order term $+\beta\widetilde{W}_{1,2}$ overwhelms the negative Laguerre defect $c_1^{\mathrm{DH}}(\tau_{\mathrm{false}}) < 0$ and renders $K^{\mathrm{DH}}_{\alpha,\beta}$ blind to actual off-line zeros.

Theorem 1.1 (Main Structural and Blind-Spot Results)**.** *Let $F(z)$ be a real-entire function ($\overline{F(\bar z)} = F(z)$), let $z = t - \mathrm{i}d \in \mathbb{C}$ with $d \in \mathbb{R}\setminus\{0\}$ and $t \in \mathbb{R}$, and set $E_{\alpha,\beta}(z) := F(z) - \mathrm{i}\alpha F'(z) - \beta F''(z)$ and $E^{\#}_{\alpha,\beta}(z) := \overline{E_{\alpha,\beta}(\bar z)} = F(z) + \mathrm{i}\alpha F'(z) - \beta F''(z)$ for $\alpha, \beta \in \mathbb{R}$. Then:*

(i) ***Cross-Term-Free Kernel Splitting and Cauchy–Pick Decomposition.*** *In $|E_{\alpha,\beta}(t - \mathrm{i}d)|^2 - |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)|^2$, all three squared moduli $|F|^2, \alpha^2|F'|^2, \beta^2|F''|^2$ and the non-adjacent cross-term $-2\beta\mathrm{Re}(\overline{F}F'')$ cancel identically, yielding*

$$K_{\alpha,\beta}(t - \mathrm{i}d,\, t - \mathrm{i}d) := \frac{|E_{\alpha,\beta}(t - \mathrm{i}d)|^2 - |E^{\#}_{\alpha,\beta}(t - \mathrm{i}d)|^2}{4\pi d} \equiv \frac{\alpha}{\pi}\,\widetilde{W}_{0,1}(d,t) + \frac{\alpha\beta}{\pi}\,\widetilde{W}_{1,2}(d,t), \tag{1.3}$$

where $\widetilde{W}_{k,k+1}(d,t) := \frac{1}{2d}\partial_d |F^{(k)}(t - \mathrm{i}d)|^2$ ($k = 0, 1$). For any nodes $z_1, \dots, z_M \in \mathbb{C}_-$, the $M \times M$ Cauchy–Pick matrix $[\mathbf{G}_{\alpha,\beta}]_{j,k} = K_{\alpha,\beta}(z_j, z_k)$ splits as $\mathbf{G}_{\alpha,\beta} \equiv \alpha\mathbf{G}^{(0,1)} + \alpha\beta\mathbf{G}^{(1,2)}$.

(ii) ***Critical-Line Lehmer Lifting at Order d^0 and Square Interlocking for Ξ.*** *Specializing to $F = \Xi$ and $\alpha, \beta > 0$, taking $d \to 0^+$ yields $\pi K_{\alpha,\beta}(t,t) \equiv \alpha c_1(t) + \alpha\beta c_1^{(1)}(t)$ with $c_1 = (\Xi')^2 - \Xi\Xi''$ and $c_1^{(1)} = (\Xi'')^2 - \Xi'\Xi'''$. At the local extremum t_{mid} ($\Xi'(t_{\mathrm{mid}}) = 0$) between a simple Lehmer pair $t_\pm = t_{\mathrm{mid}} \pm \delta$, $-\Xi'(t_{\mathrm{mid}})\Xi'''(t_{\mathrm{mid}}) = 0$ vanishes identically, leaving $\pi K_{\alpha,\beta}(t_{\mathrm{mid}}, t_{\mathrm{mid}}) = -\alpha\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) + \alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 \geq \alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 > 0$.*

(iii) ***Radial 4D-Laplace Coupling, Canonical Wigner Operator, and Hurwitz Chains.*** *The derivative Wronskian satisfies $\widetilde{W}_{1,2} \equiv \frac{1}{4}(\partial_d^2 + \frac{2}{d}\partial_d + \partial_t^2)\widetilde{W}_{0,1}$ and $c^{(1)}_{2m-1}(t) \equiv \frac{1}{4}(2m(2m+1)c_{2m+1}(t) + c''_{2m-1}(t))$ ($m \geq 1$). In canonical Wigner coordinates, $W_{1,2}(d,t) = 16\int_0^\infty y\sinh(2dy)\, I^{(1)}(y,t)\,\mathrm{d}y$ with $I^{(1)}(y,t) = (y^2 + \frac{1}{4}\partial_t^2)I(y,t)$, giving Riemann–Siegel diagonal $\mathcal{D}_{1,2}(0,t) \sim \frac{1}{40}(\ln\frac{t}{2\pi})^5$ and horizon pair amplitude $|C^{(1,2)}_{N-1,N}| = \mathcal{O}(N(t)^{-5}) = \mathcal{O}(t^{-5/2})$. For any degree-$M$ Hurwitz polynomial $P_M(X) = \sum_{r=0}^M \alpha_r X^r$ ($\alpha_0 = 1, \alpha_r > 0$), $E_{\boldsymbol{\alpha}}(z) := P_M(-\mathrm{i}\partial_z)\Xi(z)$ satisfies $\pi K_{\boldsymbol{\alpha}}(t_0,t_0) = +\alpha_{m-1}\alpha_m(\Xi^{(m)}(t_0))^2 + \mathcal{O}(\alpha_{>m})$ at a multiplicity-m zero ($1 \leq m \leq M$).*

(iv) ***Davenport–Heilbronn Blind Spot and Zero Coalescence.*** *For the self-dual Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$, which has an off-line zero at $z_* \approx 85.699348 - 0.308517\mathrm{i}$ and a critical-line Laguerre defect $c_1^{\mathrm{DH}}(\tau_{\mathrm{false}}) < 0$ at $\tau_{\mathrm{false}} \approx 85.747201$, the second-order kernel satisfies $\pi K^{\mathrm{DH}}_{1,\beta}(\tau_{\mathrm{false}}, \tau_{\mathrm{false}}) > 0$ for all $\beta > \beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}}) \approx 0.05349534$ and $\pi K^{\mathrm{DH}}_{1,\beta}(z_*, z_*) > 0$ for all $\beta > 0$. Locally, the first conjugate zero pair of $\Xi_{\mathrm{DH}}(z) - \beta\Xi''_{\mathrm{DH}}(z)$ coalesces onto $\mathbb{R}$ at $\beta^{(1)}_{\mathrm{coal}} \approx 0.04467500$, whereas globally $\beta^*(\Xi_{\mathrm{DH}}) = +\infty$ because $\Xi_{\mathrm{DH}} - \beta\Xi''_{\mathrm{DH}}$ retains infinitely many zeros in $\sigma > 1$ for every fixed $\beta \in \mathbb{R}$.*

Representative 60-decimal benchmark constants from Theorem 1.1 (for $\alpha = 1, \beta = \frac{1}{4}$) are

$$\begin{aligned} K_{1,1/4}(\gamma_1,\gamma_1) &= 6.905897\times 10^{-7}, & c_1^{(1)}(\tau_2) &= \big(\Xi''(\tau_2)\big)^2 = 6.741923\times 10^{-11},\\ \beta^{(1)}_{\mathrm{coal}}(\Xi_{\mathrm{DH}}) &= 0.0446750013, & \beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}}) &= 0.0534953385,\\ \pi K^{\mathrm{DH}}_{1,1/4}(\tau_{\mathrm{false}},\tau_{\mathrm{false}}) &= +2.378177\times 10^{-56}, & \frac{|C^{(1,2)}_{N-1,N}(10^8)|}{\mathcal{D}_{1,2}(10^8)} &= 9.4585\times 10^{-23}. \end{aligned} \tag{1.4}$$

1.1. **Proof overview.** Our analysis combines four components across Sections 2–5:

(1) *Hermitian polarization and zero-locus interpretation (Section 2).* Grouping $E_{\alpha,\beta}(z) = A_\beta(z) - \mathrm{i}\alpha B(z)$ into its real-entire parts $A_\beta := F - \beta F''$ and $B := F'$, polarization cancels $A_\beta(z_1)\overline{A_\beta(z_2)} + \alpha^2 B(z_1)\overline{B(z_2)}$ identically, proving (1.3) while showing directly why $E_{\alpha,\beta} \in \overline{\mathcal{HB}}$ tests real zeros of A_β rather than of F.

(2) *Order-d^0 pure-square collapse at critical-line extrema (Section 3).* Expanding $\widetilde{W}_{1,2}(d,t) = \sum_{m=1}^{\infty} c^{(1)}_{2m-1}(t) d^{2m-2}$, the identity $\Xi'(t_{\mathrm{mid}}) = 0$ kills $-\Xi'\Xi'''$ at Lehmer midpoints, leaving $+\alpha\beta(\Xi''(t_{\mathrm{mid}}))^2 > 0$.

(3) *Subharmonic Laplace commutation and canonical Wigner calculus (Section 4).* Applying $\frac{1}{2}\partial_d$ to $(\partial_d^2 + \partial_t^2)|F(t-\mathrm{i}d)|^2 = 4|F'(t-\mathrm{i}d)|^2$ yields $W_{1,2} \equiv \frac{1}{4}(\partial_d^2 + \partial_t^2) W_{0,1}$, from which the radial 4D-Laplace PDE, dipole recursion, canonical Wigner operator $I^{(1)} = (y^2 + \frac{1}{4}\partial_t^2) I$, and Bézout–Hurwitz parity alternation follow.

(4) *High-precision Davenport–Heilbronn evaluation and β^* threshold analysis (Section 5).* Evaluating Ξ_{DH} via Hurwitz zeta functions at 60-decimal precision locates $\beta^{(1)}_{\mathrm{coal}}$ and $\beta^{(1)}_{\mathrm{blind}}$, motivating the differential Newman-analogue threshold $\beta^*(F)$.

1.2. **Notation and conventions.** Throughout, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and $t \in \mathbb{R}$, and $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$ (so $z \in \mathbb{C}_-$ corresponds to $d = -\mathrm{Im}(z) > 0$). Riemann's completed zeta function is $\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ and $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z) = R(d,t) + \mathrm{i}\, I(d,t)$. For any real-entire function $F(z)$ and integer $k \geq 0$, $\partial_d F^{(k)}(t - \mathrm{i}d) = -\mathrm{i}\, F^{(k+1)}(t - \mathrm{i}d)$, and the canonical k-th transverse Wronskian and parity-reduced Wronskian are

$$W_{k,k+1}(d,t) := \frac{1}{2}\partial_d \big|F^{(k)}(t-\mathrm{i}d)\big|^2 = \mathrm{Im}\Big(\overline{F^{(k)}(t-\mathrm{i}d)}\, F^{(k+1)}(t-\mathrm{i}d)\Big), \quad \widetilde{W}_{k,k+1}(d,t) := \frac{W_{k,k+1}(d,t)}{d}. \tag{1.5}$$

For $F = \Xi$, we enforce the canonical Fourier–Wigner normalization across the series:

$$\Xi(z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u, \qquad W_{0,1}(d,t) = 16\int_0^\infty y\sinh(2dy)\, I(y,t)\,\mathrm{d}y, \tag{1.6}$$

where $\Phi(u) := \sum_{n=1}^{\infty}(2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u})\exp(-\pi n^2 \mathrm{e}^{4u})$ and $I(y,t) := 2\int_0^\infty \Phi(\frac{y+r}{2})\Phi(\frac{y-r}{2})\cos(2tr)\,\mathrm{d}r$, so that $\widetilde{W}_{0,1}(d,t) = 32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)\, I(y,t)\,\mathrm{d}y$ with $\mathrm{sinc}_{\mathrm{hyp}}(x) := \frac{\sinh x}{x}$. All numerical benchmarks and interval enclosures are produced by `scripts/verify_paper13_second_order_debranges.py` at 60-decimal, 550-decimal, and 35-decimal interval (`mpmath.iv`) precision.

2. The second-order de Branges–Nevanlinna kernel and Cauchy–Pick splitting

Recall that an entire function $E(z)$ belongs to the Hermite–Biehler class $\mathcal{HB}$ (with respect to the lower half-plane $\mathbb{C}_- = \{w = t - \mathrm{i}d : d > 0\}$) if $E(z)$ has no zeros in $\mathbb{C}_-$ and $|E(w)| > |E^{\#}(w)|$ for all $w \in \mathbb{C}_-$, where $E^{\#}(z) := \overline{E(\bar{z})}$ [6]; the closure $\overline{\mathcal{HB}}$ allows $|E(w)| \geq |E^{\#}(w)|$ and real zeros common to E and $E^{\#}$. Whether or not E belongs to $\mathcal{HB}$, the Hermitian difference quotient

$K_E(z_1, z_2) := \frac{E(z_1)\overline{E(z_2)} - E^\#(z_1)\overline{E^\#(z_2)}}{2\pi\mathrm{i}(z_1 - \bar{z}_2)}$ is a well-defined real-analytic kernel, which is a positive-definite reproducing kernel on $\mathbb{C}_-$ if and only if $E \in \overline{\mathcal{HB}}$ without identically vanishing modulus difference.

Theorem 2.1 (Second-Order De Branges–Nevanlinna Kernel Splitting)**.** *Let $F(z)$ be any real-entire function. For all $\alpha, \beta \in \mathbb{R}$ and every $w = t - \mathrm{i}d \in \mathbb{C}$ with $d \neq 0$, let $E_{\alpha,\beta}(z) := F(z) - \mathrm{i}\alpha F'(z) - \beta F''(z)$ and $E^\#_{\alpha,\beta}(z) := F(z) + \mathrm{i}\alpha F'(z) - \beta F''(z)$. Then in $|E_{\alpha,\beta}(w)|^2 - |E^\#_{\alpha,\beta}(w)|^2$, the three squared moduli $|F(w)|^2$, $\alpha^2|F'(w)|^2$, $\beta^2|F''(w)|^2$ and the $(0,2)$ cross-term $-2\beta\mathrm{Re}(\overline{F(w)}F''(w))$ all cancel identically, yielding*

$$\boxed{K_{\alpha,\beta}(t - \mathrm{i}d,\, t - \mathrm{i}d) = \frac{|E_{\alpha,\beta}(t - \mathrm{i}d)|^2 - |E^\#_{\alpha,\beta}(t - \mathrm{i}d)|^2}{4\pi d} \equiv \frac{\alpha}{\pi}\,\widetilde{W}_{0,1}(d,t) + \frac{\alpha\beta}{\pi}\,\widetilde{W}_{1,2}(d,t).} \tag{2.1}$$

Consequently, for $\alpha > 0$ and $\beta > 0$, the strict pointwise inequality $|E_{\alpha,\beta}(t - \mathrm{i}d)| > |E^\#_{\alpha,\beta}(t - \mathrm{i}d)|$ holds at $w = t - \mathrm{i}d$ ($d > 0$) if and only if $\widetilde{W}_{0,1}(d,t) + \beta\,\widetilde{W}_{1,2}(d,t) > 0$, with exact ratio defect

$$1 - \left|\frac{E^\#_{\alpha,\beta}(t - \mathrm{i}d)}{E_{\alpha,\beta}(t - \mathrm{i}d)}\right|^2 \equiv \frac{4\alpha d\Big(\widetilde{W}_{0,1}(d,t) + \beta\,\widetilde{W}_{1,2}(d,t)\Big)}{\big|E_{\alpha,\beta}(t - \mathrm{i}d)\big|^2} \qquad \big(E_{\alpha,\beta}(t - \mathrm{i}d) \neq 0\big). \tag{2.2}$$

Proof. Set $A_\beta(z) := F(z) - \beta\,F''(z)$ and $B(z) := F'(z)$. Because F is real-entire and $\alpha, \beta \in \mathbb{R}$, both A_β and B are real-entire, so

$$E_{\alpha,\beta}(z) = A_\beta(z) - \mathrm{i}\alpha\,B(z), \qquad E^\#_{\alpha,\beta}(z) = A_\beta(z) + \mathrm{i}\alpha\,B(z). \tag{2.3}$$

Expanding the squared moduli at $w = t - \mathrm{i}d$ gives

$$\big|E_{\alpha,\beta}(w)\big|^2 = \big|A_\beta(w)\big|^2 + \alpha^2|B(w)|^2 + 2\alpha\,\mathrm{Im}\Big(\overline{A_\beta(w)}\,B(w)\Big), \tag{2.4}$$

$$\big|E^\#_{\alpha,\beta}(w)\big|^2 = \big|A_\beta(w)\big|^2 + \alpha^2|B(w)|^2 - 2\alpha\,\mathrm{Im}\Big(\overline{A_\beta(w)}\,B(w)\Big). \tag{2.5}$$

The common sum $|A_\beta(w)|^2 + \alpha^2|B(w)|^2 = |F(w)|^2 + \alpha^2|F'(w)|^2 + \beta^2|F''(w)|^2 - 2\beta\,\mathrm{Re}(\overline{F(w)}F''(w))$ cancels identically upon subtracting (2.5) from (2.4). Using $\mathrm{Im}(\overline{F''(w)}F'(w)) = -\mathrm{Im}(\overline{F'(w)}F''(w))$ in the remaining cross-term gives

$$\begin{aligned}\big|E_{\alpha,\beta}(w)\big|^2 - \big|E^\#_{\alpha,\beta}(w)\big|^2 &= 4\alpha\,\mathrm{Im}\Big(\overline{F(w)}\,F'(w)\Big) + 4\alpha\beta\,\mathrm{Im}\Big(\overline{F'(w)}\,F''(w)\Big)\\ &= 4\alpha d\,\widetilde{W}_{0,1}(d,t) + 4\alpha\beta d\,\widetilde{W}_{1,2}(d,t).\end{aligned}$$

Dividing by $4\pi d$ and by $|E_{\alpha,\beta}(w)|^2$ proves (2.1) and (2.2). Alternatively, writing $F(t - \mathrm{i}d) = R + \mathrm{i}I$ with $\partial_d F = -\mathrm{i}F'$ and $\partial_d^2 F = -F''$ gives $|E_{\alpha,\beta}|^2 - |E^\#_{\alpha,\beta}|^2 = 2\alpha\partial_d(R^2 + I^2) + 2\alpha\beta\partial_d((\partial_d R)^2 + (\partial_d I)^2)$, reaching the same result by the Cauchy–Riemann equations. □

Proposition 2.2 (Off-Diagonal Cauchy–Pick Splitting and Uniqueness of Degree Two)**.** *For any real-entire $F(z)$ and any $z_1, z_2 \in \mathbb{C}$ with $z_1 \neq \bar{z}_2$, the two-point Hermitian kernel*

$$K_{\alpha,\beta}(z_1, z_2) := \frac{E_{\alpha,\beta}(z_1)\overline{E_{\alpha,\beta}(z_2)} - E^\#_{\alpha,\beta}(z_1)\overline{E^\#_{\alpha,\beta}(z_2)}}{2\pi\mathrm{i}(z_1 - \bar{z}_2)} \tag{2.6}$$

splits identically as

$$K_{\alpha,\beta}(z_1, z_2) \equiv \frac{\alpha}{\pi}\,\frac{F(z_1)\overline{F'(z_2)} - F'(z_1)\overline{F(z_2)}}{z_1 - \bar{z}_2} + \frac{\alpha\beta}{\pi}\,\frac{F'(z_1)\overline{F''(z_2)} - F''(z_1)\overline{F'(z_2)}}{z_1 - \bar{z}_2}. \tag{2.7}$$

Hence for any tuple $z_1, \dots, z_M \in \mathbb{C}_-$, the $M \times M$ Cauchy–Pick matrix $[\mathbf{G}_{\alpha,\beta}]_{j,k} := K_{\alpha,\beta}(z_j, z_k)$ satisfies $\mathbf{G}_{\alpha,\beta} \equiv \alpha\mathbf{G}^{(0,1)} + \alpha\beta\mathbf{G}^{(1,2)}$ with zero non-adjacent jump terms, and degree $M_{\mathrm{deg}} = 2$ is the unique polynomial order above $M_{\mathrm{deg}} = 1$ with this property (Appendix A.2).

TABLE 1. Sixty-decimal verification of the second-order kernel splitting (2.1) for $F = \Xi$ at $\alpha = 1, \beta = \frac{1}{4}$ across twelve benchmark points (d, t), where $\gamma_1 \approx 14.134725$, $\tau_1 \approx 15.585709$, $\nu_1 \approx 17.033801$, $\gamma_2 \approx 21.022040$, and $\tau_2 \approx 22.097977$.

Point (d,t)	$K_{1,1/4}(t-\mathrm{i}d, t-\mathrm{i}d)$	$\widetilde{W}_{0,1}(d,t)$	$\widetilde{W}_{1,2}(d,t)$	60-dps Err	Status
$(0.005,\ \gamma_1)$	$6.90592051742077 \times 10^{-7}$	$1.91191854986035 \times 10^{-6}$	$1.03056146608022 \times 10^{-6}$	1.21×10^{-59}	VERIFIED
$(0.050,\ \gamma_1)$	$6.90820290250873 \times 10^{-7}$	$1.91255916890467 \times 10^{-6}$	$1.03086711959297 \times 10^{-6}$	2.15×10^{-61}	VERIFIED
$(0.250,\ \gamma_1)$	$6.96368736240442 \times 10^{-7}$	$1.92813270200324 \times 10^{-6}$	$1.03829681583739 \times 10^{-6}$	0.00×10^{0}	VERIFIED
$(0.500,\ \gamma_1)$	$7.13899321516852 \times 10^{-7}$	$1.97734020580769 \times 10^{-6}$	$1.06176263228955 \times 10^{-6}$	2.08×10^{-61}	VERIFIED
$(0.150,\ \tau_1)$	$1.31041685077785 \times 10^{-7}$	$3.61120508402209 \times 10^{-7}$	$2.02236347008751 \times 10^{-7}$	4.25×10^{-61}	VERIFIED
$(0.150,\ \tau_2)$	$2.58292245878322 \times 10^{-11}$	$6.41850641538036 \times 10^{-11}$	$6.78393522370042 \times 10^{-11}$	5.26×10^{-61}	VERIFIED
$(0.100,\ \gamma_2)$	$1.25357143922851 \times 10^{-10}$	$3.15930366649816 \times 10^{-10}$	$3.11562863092838 \times 10^{-10}$	8.67×10^{-61}	VERIFIED
$(0.200,\ \nu_1)$	$2.51163606661675 \times 10^{-8}$	$6.90423549750514 \times 10^{-8}$	$3.94520767147690 \times 10^{-8}$	1.38×10^{-60}	VERIFIED
$(0,\ \gamma_1)\ (d \to 0^+)$	$6.90589746558832 \times 10^{-7}$	$1.91191207968303 \times 10^{-6}$	$1.03055837900254 \times 10^{-6}$	0.0 (exact)	VERIFIED
$(0,\ \tau_1)\ (d \to 0^+)$	$1.30588008612953 \times 10^{-7}$	$3.59864492317570 \times 10^{-7}$	$2.01559344751214 \times 10^{-7}$	0.0 (exact)	VERIFIED
$(0,\ \tau_2)\ (d \to 0^+)$	$2.56404827953347 \times 10^{-11}$	$6.36971456177266 \times 10^{-11}$	$6.74192270663692 \times 10^{-11}$	0.0 (exact)	VERIFIED
$(0,\ \nu_1)\ (d \to 0^+)$	$2.49521132719322 \times 10^{-8}$	$6.85931818774240 \times 10^{-8}$	$3.91847754768744 \times 10^{-8}$	0.0 (exact)	VERIFIED

Proof. Using (2.3), $E_{\alpha,\beta}(z_1)\overline{E_{\alpha,\beta}(z_2)} - E^{\#}_{\alpha,\beta}(z_1)\overline{E^{\#}_{\alpha,\beta}(z_2)} = 2\mathrm{i}\alpha(A_\beta(z_1)\overline{B(z_2)} - B(z_1)\overline{A_\beta(z_2)})$. Substituting $A_\beta = F - \beta F''$ and $B = F'$ and dividing by $2\pi\mathrm{i}(z_1 - \bar{z}_2)$ yields (2.7). □

Remark 2.3 (Crucial Distinction Between $E_{\alpha,\beta} \in \overline{\mathcal{HB}}$ and the Riemann Hypothesis). Decomposition (2.3) makes transparent what $E_{\alpha,\beta} \in \overline{\mathcal{HB}}$ actually means zero-theoretically. By Hermite–Biehler's theorem [6, Chapter 1], an entire function $E = A - \mathrm{i}\alpha B$ ($\alpha > 0$) with real-entire components A, B belongs to $\overline{\mathcal{HB}}$ (with $|E(w)| > |E^{\#}(w)|$ on $\mathbb{C}_-$) if and only if all zeros of A and B are real and interlace with $A'(x)B(x) - A(x)B'(x) \le 0$ (or the sign-reversed orientation on $\mathbb{C}_-$). For $E_{\alpha,\beta}$, the even-order component is $A_\beta(z) = F(z) - \beta F''(z)$, *not* $F(z)$. By the Jensen–Pólya theorem on real-rooted differential operators applied to entire functions of genus 0 or 1 [14, 21] (see also Csordas, Smith, and Varga [4]), if F belongs to the Laguerre–Pólya class $\mathcal{LP}$, then $(1 - \beta\partial_z^2)F = (1 - \mathrm{i}\sqrt{\beta}\partial_z)(1 + \mathrm{i}\sqrt{\beta}\partial_z)F$ also belongs to $\mathcal{LP}$ for every $\beta > 0$, because the polynomial $P(X) = 1 - \beta X^2$ applied to ∂_z has real roots $X = \pm\beta^{-1/2}$. Crucially, the converse is *false* for any fixed $\beta > 0$: applying $1 - \beta\partial_z^2$ pulls sufficiently narrow conjugate pairs of nonreal zeros of F onto the real axis (see Section 5 for an explicit example on Ξ_{DH}). Thus, for $F = \Xi$:

$$\mathrm{RH} \implies E_{\alpha,\beta} \in \overline{\mathcal{HB}} \quad (\forall\, \alpha > 0,\ \beta \ge 0), \qquad \text{whereas for fixed } \beta > 0, \quad E_{\alpha,\beta} \in \overline{\mathcal{HB}} \not\Longrightarrow \mathrm{RH}.$$

Equivalence to the Riemann Hypothesis is recovered only in the limit $\beta \to 0^+$ (i.e., $E_{\alpha,\beta} \in \overline{\mathcal{HB}}$ for *all* $\beta > 0$).

3. Critical-line Lehmer near-coalescence lifting and square interlocking

Specializing to $F = \Xi$, because $\Xi'(z)$ is real-entire (with odd parity $\Xi'(-z) = -\Xi'(z)$), $|\Xi'(t - \mathrm{i}d)|^2 = \Xi'(t - \mathrm{i}d)\Xi'(t + \mathrm{i}d)$ is an even real-analytic function of d, so $W_{1,2}(d,t)$ is odd in d. Dividing by d yields the globally convergent power series

$$\widetilde{W}_{1,2}(d,t) = \sum_{m=1}^{\infty} c^{(1)}_{2m-1}(t)\, d^{2m-2} = c^{(1)}_1(t) + c^{(1)}_3(t)\, d^2 + c^{(1)}_5(t)\, d^4 + \mathcal{O}(d^6), \tag{3.1}$$

where the derivative transverse Taylor moments $c^{(1)}_{2m-1}(t)$ are obtained by shifting $\Xi^{(j)} \mapsto \Xi^{(j+1)}$ in the standard Taylor expansion of $\frac{1}{2d}\partial_d|\Xi(t - \mathrm{i}d)|^2$:

$$c^{(1)}_{2m-1}(t) := \frac{\partial_d^{2m-1} W_{1,2}(0,t)}{(2m-1)!} = m \sum_{j=0}^{2m} \frac{(-1)^{m+j}}{j!\,(2m-j)!}\, \Xi^{(j+1)}(t)\, \Xi^{(2m-j+1)}(t). \tag{3.2}$$

In particular, for $m = 1$ and $m = 2$,

$$c_1^{(1)}(t) = \big(\Xi''(t)\big)^2 - \Xi'(t)\Xi'''(t), \qquad c_3^{(1)}(t) = \frac{1}{6}\Big(3\big(\Xi'''(t)\big)^2 - 4\Xi''(t)\Xi^{(4)}(t) + \Xi'(t)\Xi^{(5)}(t)\Big). \quad (3.3)$$

Theorem 3.1 (Critical-Line Lehmer Lifting at Order d^0 and Square Interlocking)**.** *Fix any coupling parameters $\alpha > 0$ and $\beta > 0$. Then for $F = \Xi$:*

(i) ***Critical-line limit and order-d^0 lifting at double zeros.*** *As $d \to 0^+$,*

$$\pi\, K_{\alpha,\beta}(t,t) = \lim_{d\to 0^+} \pi\, K_{\alpha,\beta}(t-\mathrm{i}d,\, t-\mathrm{i}d) \equiv \alpha\Big(\big(\Xi'(t)\big)^2 - \Xi(t)\Xi''(t)\Big) + \alpha\beta\Big(\big(\Xi''(t)\big)^2 - \Xi'(t)\Xi'''(t)\Big). \quad (3.4)$$

At any hypothetical double real zero $t_0 \in \mathbb{R}$ ($\Xi(t_0) = \Xi'(t_0) = 0$, $\Xi''(t_0) \neq 0$), $c_1(t_0) = 0$ vanishes whereas $c_1^{(1)}(t_0) = (\Xi''(t_0))^2 > 0$, so $\pi K_{\alpha,\beta}(t_0,t_0) = +\alpha\beta(\Xi''(t_0))^2 > 0$. More generally, at a real zero t_0 of multiplicity $m \geq 2$, the leading transverse power shifts down from d^{2m-2} to d^{2m-4}:

$$\pi\, K_{\alpha,\beta}(t_0 - \mathrm{i}d,\, t_0 - \mathrm{i}d) = \alpha\beta\, \frac{m-1}{\big((m-1)!\big)^2}\Big(\Xi^{(m)}(t_0)\Big)^2 d^{2m-4} + \mathcal{O}(d^{2m-2}) > 0 \qquad (\forall\, m \geq 2). \quad (3.5)$$

(ii) ***Order-d^0 floor at simple Lehmer midpoints.*** *Let $t_\pm = t_{\mathrm{mid}} \pm \delta$ ($\delta > 0$) be consecutive simple real zeros of $\Xi(t)$, and let $t_{\mathrm{mid}} \in (t_-, t_+)$ be an intermediate local extremum with $\Xi'(t_{\mathrm{mid}}) = 0$ and $-\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) > 0$. Because $\Xi'(t_{\mathrm{mid}}) = 0$, the product $-\Xi'(t_{\mathrm{mid}})\Xi'''(t_{\mathrm{mid}}) = 0$ vanishes identically in (3.4), giving*

$$\pi\, K_{\alpha,\beta}(t_{\mathrm{mid}}, t_{\mathrm{mid}}) \equiv -\alpha\, \Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) + \alpha\beta\big(\Xi''(t_{\mathrm{mid}})\big)^2 \geq +\alpha\beta\big(\Xi''(t_{\mathrm{mid}})\big)^2 > 0, \quad (3.6)$$

and locally around $(0, t_{\mathrm{mid}})$,

$$\pi\, K_{\alpha,\beta}(t_{\mathrm{mid}}+\tau-\mathrm{i}d,\, t_{\mathrm{mid}}+\tau-\mathrm{i}d) = \alpha\beta\big(\Xi''(t_{\mathrm{mid}})\big)^2 + \frac{\alpha}{2}\big(\Xi''(t_{\mathrm{mid}})\big)^2\big(\delta^2+\tau^2+d^2\big) + \mathcal{O}\big(\beta|\tau| + (\delta+|\tau|+d)^3\big). \quad (3.7)$$

(iii) ***Interlocking of pure squares along $d = 0$.*** *Across simple zeros γ_k ($\Xi(\gamma_k) = 0$), local extrema τ_k ($\Xi'(\tau_k) = 0$), and inflection points ν_k ($\Xi''(\nu_k) = 0$), at least one component reduces to a pure square:*

$$\begin{aligned} \pi\, K_{\alpha,\beta}(\gamma_k, \gamma_k) &= \alpha\big(\Xi'(\gamma_k)\big)^2 + \alpha\beta\, c_1^{(1)}(\gamma_k), \\ \pi\, K_{\alpha,\beta}(\tau_k, \tau_k) &= -\alpha\, \Xi(\tau_k)\Xi''(\tau_k) + \alpha\beta\big(\Xi''(\tau_k)\big)^2, \\ \pi\, K_{\alpha,\beta}(\nu_k, \nu_k) &= \alpha\big(\Xi'(\nu_k)\big)^2 - \alpha\beta\, \Xi'(\nu_k)\Xi'''(\nu_k). \end{aligned} \quad (3.8)$$

Proof. (i) Taking $d \to 0^+$ in (2.1) gives (3.4). If $\Xi(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$ and $\Xi^{(m)}(t_0) \neq 0$ with $m \geq 2$, then in (3.2) every product $\Xi^{(j+1)}(t_0)\Xi^{(2n-j+1)}(t_0)$ vanishes whenever $2n + 2 < 2m$, i.e., for all $n \leq m-2$. At $n = m-1$, only the central index $j = m-1$ survives, giving $c_{2m-3}^{(1)}(t_0) = (m-1)\frac{(-1)^{2m-2}}{((m-1)!)^2}(\Xi^{(m)}(t_0))^2 > 0$ at order d^{2m-4}, whereas $\widetilde{W}_{0,1}(d, t_0)$ starts at order d^{2m-2}.

(ii)–(iii) At $\Xi'(t_{\mathrm{mid}}) = 0$, $c_1^{(1)}(t_{\mathrm{mid}}) = (\Xi''(t_{\mathrm{mid}}))^2 > 0$, while $\Xi(t_{\mathrm{mid}}) = -\frac{1}{2}\Xi''(t_{\mathrm{mid}})\delta^2 + \mathcal{O}(\delta^4)$ gives $c_1(t_{\mathrm{mid}}) = -\Xi(t_{\mathrm{mid}})\Xi''(t_{\mathrm{mid}}) = \frac{1}{2}(\Xi''(t_{\mathrm{mid}}))^2\delta^2 + \mathcal{O}(\delta^4) > 0$, establishing (3.6)–(3.7). Substituting $\Xi(\gamma_k) = 0$, $\Xi'(\tau_k) = 0$, and $\Xi''(\nu_k) = 0$ into (3.4) yields (3.8). □

4. The radial Laplace–Wigner identity, dipole recursion, and order-M Hurwitz chains

We next establish the partial differential, Wigner phase-space [27, 13], and Riemann–Siegel [24, 10] relations linking $\widetilde{W}_{1,2}$ to $\widetilde{W}_{0,1}$, together with the extension of Theorems 2.1 and 3.1 to general degree-M Hurwitz differential polynomials [14, 21].

TABLE 2. Top: 60-decimal values and 35-decimal `mpmath.iv` interval certification of $c_1(t)$, $c_1^{(1)}(t)$, and $\pi K_{1,1/4}(t,t) = c_1(t) + \frac{1}{4}c_1^{(1)}(t)$ across the first five simple zeros γ_k, extrema τ_k, and inflection points ν_k of $\Xi(t)$. Bottom: 550-decimal synthetic Lehmer deformation around $\tau_2 = 22.09797728\ldots$ down to $\Delta\gamma = 2\delta = 10^{-250}$.

Point t ($d=0$)	Coordinate t	$c_1(t)$ [Order 1]	$c_1^{(1)}(t)$ [Order 2]	$\pi K_{1,1/4}(t,t)$	35-dps `iv`
γ_1 ($\Xi=0$)	14.134725141735	$1.91191207968 \times 10^{-6}$	$1.03055837900 \times 10^{-6}$	$2.16955167443 \times 10^{-6}$	CERTIFIED
τ_1 ($\Xi'=0$)	15.585708589829	$3.59864492318 \times 10^{-7}$	$2.01559344751 \times 10^{-7}$	$4.10254328505 \times 10^{-7}$	CERTIFIED
ν_1 ($\Xi''=0$)	17.033801441961	$6.85931818774 \times 10^{-8}$	$3.91847754769 \times 10^{-8}$	$7.83893757466 \times 10^{-8}$	CERTIFIED
γ_2 ($\Xi=0$)	21.022039638772	$3.15054555331 \times 10^{-10}$	$3.10854405211 \times 10^{-10}$	$3.92768156634 \times 10^{-10}$	CERTIFIED
τ_2 ($\Xi'=0$)	22.097977280401	$6.36971456177 \times 10^{-11}$	$6.74192270664 \times 10^{-11}$	$8.05519523843 \times 10^{-11}$	CERTIFIED
ν_2 ($\Xi''=0$)	23.213767459682	$1.39820111181 \times 10^{-11}$	$1.38558400554 \times 10^{-11}$	$1.74459711319 \times 10^{-11}$	CERTIFIED
γ_3 ($\Xi=0$)	25.010857580146	$1.60074037994 \times 10^{-12}$	$1.34159315729 \times 10^{-12}$	$1.93613866926 \times 10^{-12}$	CERTIFIED
τ_3 ($\Xi'=0$)	26.272247356936	$3.58208864101 \times 10^{-13}$	$2.86291614782 \times 10^{-13}$	$4.29781767797 \times 10^{-13}$	CERTIFIED
ν_3 ($\Xi''=0$)	27.492630585603	$7.48851053471 \times 10^{-14}$	$6.40885231003 \times 10^{-14}$	$9.09072361221 \times 10^{-14}$	CERTIFIED
γ_4 ($\Xi=0$)	30.424876125860	$5.81633978842 \times 10^{-16}$	$1.01585236550 \times 10^{-15}$	$8.35597070217 \times 10^{-16}$	CERTIFIED
τ_4 ($\Xi'=0$)	31.231795871010	$1.31300145969 \times 10^{-16}$	$2.62360140193 \times 10^{-16}$	$1.96890181018 \times 10^{-16}$	CERTIFIED
ν_4 ($\Xi''=0$)	32.132928976626	$3.69775862961 \times 10^{-17}$	$5.87611378804 \times 10^{-17}$	$5.16678707662 \times 10^{-17}$	CERTIFIED
γ_5 ($\Xi=0$)	32.935061587739	$1.67213528471 \times 10^{-17}$	$1.84258253238 \times 10^{-17}$	$2.13278091780 \times 10^{-17}$	CERTIFIED
τ_5 ($\Xi'=0$)	34.193310269011	$4.75000680582 \times 10^{-18}$	$4.20481035599 \times 10^{-18}$	$5.80120939482 \times 10^{-18}$	CERTIFIED
ν_5 ($\Xi''=0$)	35.368227994114	$1.11255996557 \times 10^{-18}$	$1.09620486707 \times 10^{-18}$	$1.38661118234 \times 10^{-18}$	CERTIFIED

Lehmer Gap $\Delta\gamma = 2\delta$ at τ_2	$\pi K_1(t_{\mathrm{mid}}, t_{\mathrm{mid}})$	$c_1^{(1)}(t_{\mathrm{mid}}) = (\Xi'')^2$	$\pi K_{1,1/4}(t_{\mathrm{mid}}, t_{\mathrm{mid}})$	550-dps
$\Delta\gamma = 10^{0}$	$8.42740338330 \times 10^{-12}$	$6.74192270664 \times 10^{-11}$	$2.52822101499 \times 10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-2}$	$8.42740338330 \times 10^{-16}$	$6.74192270664 \times 10^{-11}$	$1.68556495069 \times 10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-10}$	$8.42740338330 \times 10^{-32}$	$6.74192270664 \times 10^{-11}$	$1.68548067666 \times 10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-50}$	$8.42740338330 \times 10^{-112}$	$6.74192270664 \times 10^{-11}$	$1.68548067666 \times 10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-100}$	$8.42740338330 \times 10^{-212}$	$6.74192270664 \times 10^{-11}$	$1.68548067666 \times 10^{-11}$	$>$ Floor
$\Delta\gamma = 10^{-250}$	$8.42740338330 \times 10^{-512}$	$6.74192270664 \times 10^{-11}$	$1.68548067666 \times 10^{-11}$	5.0×10^{-501}

4.1. **Double-exponential decay justification.** In the canonical logarithmic representation $\Xi(t - \mathrm{i}d) = 4\int_{-\infty}^{\infty} \Phi(u)\mathrm{e}^{2du - 2\mathrm{i}tu}\,\mathrm{d}u$, the even Jacobi–Riemann kernel $\Phi(u)$ is holomorphic in the strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$ and decays double-exponentially as $|\Phi(\xi+\mathrm{i}\eta)| \le C_0\mathrm{e}^{9|\xi|}\exp(-\pi\mathrm{e}^{4|\xi|}\cos(4\eta))$ as $|\xi| \to \infty$ whenever $|\eta| < \frac{\pi}{8}$ [20, 26]. Consequently, for every polynomial weight u^k ($k \ge 0$) and fixed $d \in \mathbb{R}$, $(-2\mathrm{i}u)^k\Phi(u)\mathrm{e}^{2du}$ belongs to the Schwartz space $\mathcal{S}(\mathbb{R})$ with super-exponential decay $\exp(-\pi\mathrm{e}^{4|u|} + 2|d||u| + \mathcal{O}(|u|))$, justifying termwise differentiation under the integral sign with respect to (d,t) and Fubini's theorem in rotated Wigner coordinates $(y,r) = (u_1+u_2, u_1-u_2)$.

Theorem 4.1 (Radial 4D-Laplace PDE, Canonical Wigner Operator, and Saddle-Boundary Scaling)**.** *Let $W_{k,k+1}(d,t)$ and $\widetilde{W}_{k,k+1}(d,t)$ be defined by (1.5). Then:*

(i) ***Global radial 4D-Laplace PDE and dipole recursion.*** *For any real-entire function $F(z)$ and all $(d,t) \in (\mathbb{R}\setminus\{0\}) \times \mathbb{R}$, $W_{1,2}(d,t) \equiv \frac{1}{4}(\partial_d^2 + \partial_t^2)W_{0,1}(d,t)$, and the parity-reduced Wronskians satisfy*

$$\widetilde{W}_{1,2}(d,t) \equiv \frac{1}{4}\left(\frac{\partial^2\widetilde{W}_{0,1}}{\partial d^2} + \frac{2}{d}\frac{\partial\widetilde{W}_{0,1}}{\partial d} + \frac{\partial^2\widetilde{W}_{0,1}}{\partial t^2}\right). \tag{4.1}$$

Equating coefficients of d^{2m-2} yields for every integer $m \ge 1$ the exact recursion

$$c_{2m-1}^{(1)}(t) \equiv \frac{1}{4}\Big(2m(2m+1)\,c_{2m+1}(t) + c_{2m-1}''(t)\Big) = \frac{m(2m+1)}{2}\,c_{2m+1}(t) + \frac{1}{4}\,c_{2m-1}''(t), \tag{4.2}$$

and in particular $c_1^{(1)}(t) \equiv \frac{3}{2}c_3(t) + \frac{1}{4}c_1''(t)$ and $c_3^{(1)}(t) \equiv 5c_5(t) + \frac{1}{4}c_3''(t)$.

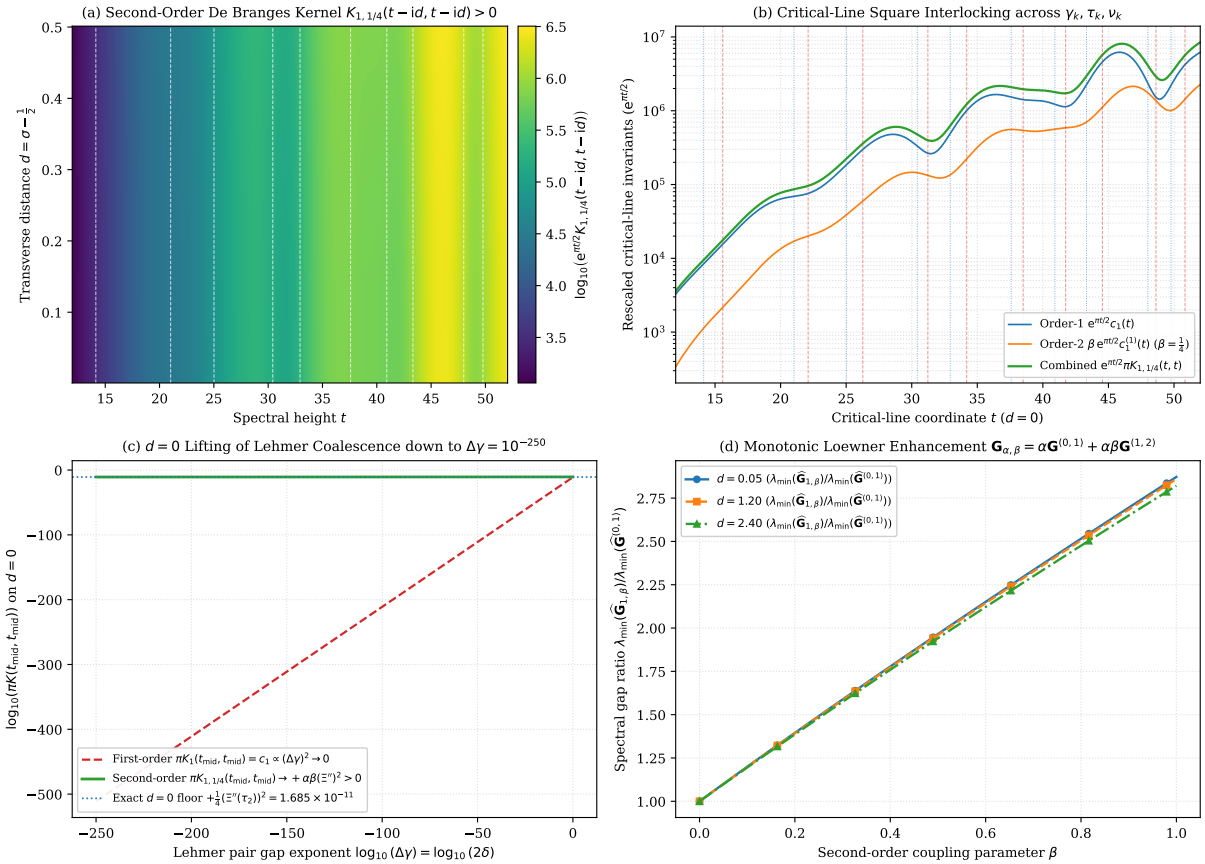


FIGURE 1. **(a)** Continuum map of $\log_{10}(e^{\pi t/2} K_{1,1/4}(t - \mathrm{i}d, t - \mathrm{i}d))$ for Ξ across $[0.002, 0.50] \times [12, 52]$ (200×500 grid). **(b)** Critical-line interlocking of $e^{\pi t/2} c_1(t)$, $\beta e^{\pi t/2} c_1^{(1)}(t)$, and $e^{\pi t/2} \pi K_{1,1/4}(t,t)$ across the first ten simple zeros γ_k, extrema τ_k, and inflection points ν_k. **(c)** 550-decimal synthetic Lehmer deformation down to $\Delta\gamma = 10^{-250}$, contrasting the $\mathcal{O}((\Delta\gamma)^2) \to 0$ decay of $\pi K_1(t_{\mathrm{mid}}, t_{\mathrm{mid}})$ against the order-d^0 floor $+\frac{1}{4}(\Xi''(\tau_2))^2 = 1.685 \times 10^{-11}$ of $\pi K_{1,1/4}(t_{\mathrm{mid}}, t_{\mathrm{mid}})$. **(d)** Loewner enhancement of the 6×6 Cauchy–Pick minimum eigenvalue ratio $\lambda_{\min}(\mathbf{G}_{1,\beta})/\lambda_{\min}(\mathbf{G}^{(0,1)})$ for $\beta \in [0, 1]$.

(ii) ***Canonical Wigner phase-space representation.*** *For $F = \Xi$ under the canonical normalization (1.6), $W_{1,2}(d,t)$ and $\widetilde{W}_{1,2}(d,t)$ satisfy*

$$W_{1,2}(d,t) = 16 \int_0^\infty y \sinh(2dy)\, I^{(1)}(y,t)\, \mathrm{d}y, \qquad \widetilde{W}_{1,2}(d,t) = 32 \int_0^\infty y^2\, \mathrm{sinc}_{\mathrm{hyp}}(2dy)\, I^{(1)}(y,t)\, \mathrm{d}y, \tag{4.3}$$

where the derivative Wigner correlation kernel $I^{(1)}(y,t)$ is given equivalently by

$$I^{(1)}(y,t) \equiv \left(y^2 + \frac{1}{4}\partial_t^2\right) I(y,t) \equiv 2 \int_0^\infty (y^2 - r^2)\, \Phi\left(\frac{y+r}{2}\right) \Phi\left(\frac{y-r}{2}\right) \cos(2tr)\, \mathrm{d}r. \tag{4.4}$$

(iii) ***Asymptotic $\frac{1}{40}(\ln \frac{t}{2\pi})^5$ diagonal and $\mathcal{O}(t^{-5/2})$ horizon-pair scaling.*** *In the Riemann–Siegel main sum up to $N(t) = \lfloor \sqrt{t/2\pi} \rfloor$ with shell frequencies $\omega_n(t) := \vartheta'(t) - \ln n \sim \frac{1}{2} \ln \frac{t}{2\pi} - \ln n$, the non-oscillatory diagonal of the rescaled derivative Wronskian $\widehat{W}_{1,2}(d,t) :=$*

$(\frac{t}{2\pi})^{1/2}\mathrm{e}^{\pi t/2}\widetilde{W}_{1,2}(d,t)$ *satisfies*

$$\mathcal{D}_{1,2}(d,t) := 4\sum_{n=1}^{N(t)} \frac{1}{n}\,\omega_n(t)^4\,\mathrm{sinc}_{\mathrm{hyp}}(2d\,\omega_n(t)) \sim \frac{1}{40}\left(\ln\frac{t}{2\pi}\right)^5 \qquad (d\to 0^+,\ t\to\infty). \tag{4.5}$$

On $d=0$, the exact off-diagonal contribution of a mode pair (m,n) $(1\le m<n\le N(t))$ with local phases $\phi_n(t) := t\ln n - \vartheta(t)$ is

$$\begin{aligned} C^{(0,1)}_{m,n}(t) &= \frac{2}{\sqrt{mn}}\Big[(\omega_m+\omega_n)^2\cos(\phi_m-\phi_n) + (\omega_m-\omega_n)^2\cos(\phi_m+\phi_n)\Big],\\ C^{(1,2)}_{m,n}(t) &= \frac{2\omega_m\omega_n}{\sqrt{mn}}\Big[(\omega_m+\omega_n)^2\cos(\phi_m-\phi_n) - (\omega_m-\omega_n)^2\cos(\phi_m+\phi_n)\Big]. \end{aligned} \tag{4.6}$$

At the saddle-point horizon $n\to N(t)$ where $\omega_n(t)=\mathcal{O}(N(t)^{-1})$, the slowest-oscillating boundary pair $(N-1,N)$ scales as $|C^{(0,1)}_{N-1,N}|=\mathcal{O}(N(t)^{-3})=\mathcal{O}(t^{-3/2})$ and $|C^{(1,2)}_{N-1,N}|=\mathcal{O}(N(t)^{-5})=\mathcal{O}(t^{-5/2})$, so $|C^{(1,2)}_{N-1,N}|/\mathcal{D}_{1,2}(0,t)=\mathcal{O}(t^{-5/2}(\ln t)^{-5})$.

Proof. (i) For any holomorphic function $F(z)=F(t-\mathrm{i}d)$, $(\partial_d^2+\partial_t^2)|F(t-\mathrm{i}d)|^2 = 4\partial_z\partial_{\bar z}(F(z)\overline{F(z)}) = 4|F'(t-\mathrm{i}d)|^2$. Applying $\frac12\partial_d$ gives $W_{1,2}\equiv\frac14(\partial_d^2+\partial_t^2)W_{0,1}$. Substituting $W_{0,1}=d\,\widetilde{W}_{0,1}$ and $\partial_d^2(d\,\widetilde{W}_{0,1}) = d\,\partial_d^2\widetilde{W}_{0,1} + 2\partial_d\widetilde{W}_{0,1}$, then dividing by $d\neq 0$, yields (4.1). Since $(\partial_d^2+\frac{2}{d}\partial_d)(d^{2m}) = 2m(2m+1)d^{2m-2}$, equating powers of d^{2m-2} yields (4.2).

(ii) By Section 4.1, $\Xi'(t-\mathrm{i}d) = 4\int_{-\infty}^{\infty}(-2\mathrm{i}u)\Phi(u)\mathrm{e}^{2du-2\mathrm{i}tu}\,\mathrm{d}u$. Squaring the modulus and changing variables to $y=u_1+u_2$, $r=u_1-u_2$ (with Jacobian $\frac12$ and $(-2\mathrm{i}u_1)(+2\mathrm{i}u_2)=4u_1u_2=y^2-r^2$) gives

$$\begin{aligned} |\Xi'(t-\mathrm{i}d)|^2 &= 8\int_{-\infty}^{\infty}\mathrm{e}^{2dy}\left(\int_{-\infty}^{\infty}(y^2-r^2)\Phi\left(\frac{y+r}{2}\right)\Phi\left(\frac{y-r}{2}\right)\cos(2tr)\,\mathrm{d}r\right)\mathrm{d}y\\ &= 16\int_0^{\infty}\cosh(2dy)\,I^{(1)}(y,t)\,\mathrm{d}y, \end{aligned}$$

since Φ is even. Differentiating via $\frac12\partial_d\cosh(2dy) = y\sinh(2dy) = 2d\,y^2\mathrm{sinc}_{\mathrm{hyp}}(2dy)$ and noting that $\frac14\partial_t^2\cos(2tr) = -r^2\cos(2tr)$ establishes (4.3)–(4.4).

(iii) For the Riemann–Siegel main sum $M(z)=\sum_{n=1}^{N}2n^{-1/2}\cos(\phi_n+\omega_n(z-t))$ at frozen frequencies $\omega_n=\omega_n(t)$, direct differentiation at $z=t$ gives $(M')^2-MM'' = \sum_{n=1}^{N}\frac{4}{n}\omega_n^2 + \sum_{1\le m<n\le N}C^{(0,1)}_{m,n}$ and $(M'')^2-M'M''' = \sum_{n=1}^{N}\frac{4}{n}\omega_n^4 + \sum_{1\le m<n\le N}C^{(1,2)}_{m,n}$ with (4.6). By Euler–Maclaurin summation with $x=\frac12\ln\frac{t}{2\pi}-\ln u\in[0,\frac12\ln\frac{t}{2\pi}]$,

$$\mathcal{D}_{1,2}(0,t)\sim 4\int_1^{\sqrt{t/2\pi}}\left(\frac12\ln\frac{t}{2\pi}-\ln u\right)^4\frac{\mathrm{d}u}{u} = 4\int_0^{\frac12\ln\frac{t}{2\pi}}x^4\,\mathrm{d}x = \frac45\left(\frac12\ln\frac{t}{2\pi}\right)^5 = \frac{1}{40}\left(\ln\frac{t}{2\pi}\right)^5.$$

Near $n=N(t)\asymp\sqrt{t/2\pi}$, $\omega_{N-1},\omega_N=\mathcal{O}(N^{-1})$ and $(mn)^{-1/2}\asymp N^{-1}$, so (4.6) gives $|C^{(0,1)}_{N-1,N}| = \mathcal{O}(N^{-1}\cdot N^{-2}) = \mathcal{O}(N^{-3}) = \mathcal{O}(t^{-3/2})$ and $|C^{(1,2)}_{N-1,N}| = \mathcal{O}(N^{-1}\cdot N^{-2}\cdot N^{-2}) = \mathcal{O}(N^{-5}) = \mathcal{O}(t^{-5/2})$. □

Theorem 4.2 (General Order-M Hurwitz Chains and Multiplicity-m Coalescence)**.** *Let $F(z)$ be real-entire, let $P_M(X)=\sum_{r=0}^{M}\alpha_r X^r = P_{\mathrm{even}}(X^2)+X\,P_{\mathrm{odd}}(X^2)$ $(\alpha_0=1,\ \alpha_r>0)$ be a real Hurwitz polynomial of degree $M\ge 1$, and define*

$$E_{\boldsymbol{\alpha}}(z) := P_M\left(-\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}z}\right)F(z) = F_{\mathrm{even}}(z) - \mathrm{i}\,F_{\mathrm{odd}}(z), \qquad E^{\#}_{\boldsymbol{\alpha}}(z) = F_{\mathrm{even}}(z)+\mathrm{i}\,F_{\mathrm{odd}}(z), \tag{4.7}$$

TABLE 3. Top: 60-decimal verification of the radial Laplace dipole recursion (4.2) for $m = 1, 2$ on Ξ. Middle: 60-decimal verification of the order-M Hurwitz coalescence law (4.9) for $P_4(X) = \prod_{j=1}^{4}(1+X/j)$ across $m = 1, 2, 3, 4$. Bottom: Exact evaluation of $\mathcal{D}_{0,1}(0,t)$, $\mathcal{D}_{1,2}(0,t) \sim \frac{1}{40}(\ln \frac{t}{2\pi})^5$, and the horizon-pair amplitude (4.6) at $(N-1, N)$ up to $t = 10^8$.

Benchmark t	$c_1^{(1)}(t)$ (Direct)	$\frac{3}{2}c_3(t) + \frac{1}{4}c_1''(t)$	$\lvert c_1^{(1)} - \text{Rec}\rvert$	$\lvert c_3^{(1)} - \text{Rec}\rvert$	Status
$\gamma_1 = 14.134725$	$1.03055837900 \times 10^{-6}$	$1.03055837900 \times 10^{-6}$	0.00×10^{0}	7.42×10^{-68}	VERIFIED
$\tau_1 = 15.585709$	$2.01559344751 \times 10^{-7}$	$2.01559344751 \times 10^{-7}$	1.85×10^{-68}	4.64×10^{-69}	VERIFIED
$\nu_1 = 17.033801$	$3.91847754769 \times 10^{-8}$	$3.91847754769 \times 10^{-8}$	0.00×10^{0}	5.80×10^{-70}	VERIFIED
$\gamma_2 = 21.022040$	$3.10854405211 \times 10^{-10}$	$3.10854405211 \times 10^{-10}$	7.24×10^{-71}	0.00×10^{0}	VERIFIED
$\tau_2 = 22.097977$	$6.74192270664 \times 10^{-11}$	$6.74192270664 \times 10^{-11}$	0.00×10^{0}	6.79×10^{-72}	VERIFIED
$\nu_2 = 23.213767$	$1.38558400554 \times 10^{-11}$	$1.38558400554 \times 10^{-11}$	1.13×10^{-72}	5.66×10^{-73}	VERIFIED
Order m	$(\alpha_{m-1},\ \alpha_m)$	$F'_{\text{even}}F_{\text{odd}} - F_{\text{even}}F'_{\text{odd}}$	$+\alpha_{m-1}\alpha_m(\Xi^{(m)})^2$	60-dps Err	Mechanism
$m = 1$ (odd)	$(1,\ \frac{25}{12})$	$3.98315016601 \times 10^{-6}$	$3.98315016601 \times 10^{-6}$	1.49×10^{-61}	$+(\Xi')^2$
$m = 2$ (even)	$(\frac{25}{12},\ \frac{35}{24})$	$7.80631459764 \times 10^{-6}$	$7.80631459764 \times 10^{-6}$	1.52×10^{-61}	$-(-1)(\Xi'')^2$
$m = 3$ (odd)	$(\frac{35}{24},\ \frac{5}{12})$	$7.52595065067 \times 10^{-7}$	$7.52595065067 \times 10^{-7}$	9.86×10^{-62}	$+(\Xi''')^2$
$m = 4$ (even)	$(\frac{5}{12},\ \frac{1}{24})$	$3.91355219285 \times 10^{-9}$	$3.91355219285 \times 10^{-9}$	0.00×10^{0}	$-(-1)(\Xi^{(4)})^2$
Height t	$N(t)$	$\mathcal{D}_{0,1}(0,t)$	$\mathcal{D}_{1,2}(0,t)$	$\lvert C^{(0,1)}_{N-1,N}\rvert/\mathcal{D}_{0,1}$	$\lvert C^{(1,2)}_{N-1,N}\rvert/\mathcal{D}_{1,2}$
$t = 10^3$	12	37.993558	196.486903	1.6035×10^{-4}	2.1260×10^{-7}
$t = 10^4$	39	100.268276	1025.904627	2.6352×10^{-6}	2.8403×10^{-10}
$t = 10^5$	126	207.751735	3510.081902	8.3812×10^{-9}	5.6760×10^{-15}
$t = 10^6$	398	372.651993	9374.799275	7.0875×10^{-10}	3.2515×10^{-16}
$t = 10^7$	1261	607.177121	21259.091331	7.4746×10^{-12}	1.1907×10^{-19}
$t = 10^8$	3989	923.535190	42909.424553	1.1624×10^{-13}	9.4585×10^{-23}

where $F_{\text{even}}(z) := \sum_{j\geq 0}(-1)^j \alpha_{2j} F^{(2j)}(z)$ *and* $F_{\text{odd}}(z) := \sum_{k\geq 0}(-1)^k \alpha_{2k+1} F^{(2k+1)}(z)$. *Then for all* $w = t - \mathrm{i}d$ *(*$d \neq 0$*),* $K_{\boldsymbol{\alpha}}(w,w) := \frac{|E_{\boldsymbol{\alpha}}(w)|^2 - |E^{\#}_{\boldsymbol{\alpha}}(w)|^2}{4\pi d} \equiv \frac{1}{\pi d}\mathrm{Im}(\overline{F_{\text{even}}(w)}F_{\text{odd}}(w))$*, with critical-line limit*

$$\begin{aligned} \pi\, K_{\boldsymbol{\alpha}}(t,t) &\equiv F'_{\text{even}}(t)\, F_{\text{odd}}(t) - F_{\text{even}}(t)\, F'_{\text{odd}}(t) \\ &= \sum_{j,k\geq 0} (-1)^{j+k}\alpha_{2j}\alpha_{2k+1}\Big(F^{(2j+1)}(t)F^{(2k+1)}(t) - F^{(2j)}(t)F^{(2k+2)}(t)\Big). \end{aligned} \tag{4.8}$$

If $t_0 \in \mathbb{R}$ *is a real zero of* $F(t)$ *of multiplicity* m *(*$1 \leq m \leq M$*), then for both odd and even* m*,*

$$\begin{aligned} \pi\, K_{\boldsymbol{\alpha}}(t_0,t_0) &= +\alpha_{m-1}\alpha_m\Big(F^{(m)}(t_0)\Big)^2 \\ &\quad + \sum_{\substack{2j+1\geq m,\ 2k+1\geq m \\ j+k>m-1}} (-1)^{j+k}\alpha_{2j}\alpha_{2k+1}\Big(F^{(2j+1)}(t_0)F^{(2k+1)}(t_0) - F^{(2j)}(t_0)F^{(2k+2)}(t_0)\Big), \end{aligned} \tag{4.9}$$

which for $M = m$ *collapses to the single pure square* $\pi K_{\boldsymbol{\alpha}}(t_0,t_0) \equiv +\alpha_{m-1}\alpha_m(F^{(m)}(t_0))^2 > 0$.

Proof. Since $(-\mathrm{i}\partial_z)^{2j} = (-1)^j\partial_z^{2j}$ and $(-\mathrm{i}\partial_z)^{2k+1} = -\mathrm{i}(-1)^k\partial_z^{2k+1}$, F_{even} and F_{odd} are real-entire, and polarization gives $K_{\boldsymbol{\alpha}}(w,w) = \frac{1}{\pi d}\mathrm{Im}(\overline{F_{\text{even}}(w)}F_{\text{odd}}(w))$ and (4.8) as $d \to 0^+$. At a zero t_0 of multiplicity $m \in \{1,\dots,M\}$, if $m = 2r+1$ is odd, the unique minimal non-zero product in (4.8) occurs at $j = k = r$ in $+F^{(2j+1)}F^{(2k+1)}$ with sign $(-1)^{2r} = +1$, giving $+\alpha_{2r}\alpha_{2r+1}(F^{(m)}(t_0))^2$; if $m = 2r$ is even, the unique minimal non-zero product occurs at $j = r, k = r-1$ in $-F^{(2j)}F^{(2k+2)}$ with sign $(-1)^{2r-1}(-1) = +1$, again giving $+\alpha_{2r-1}\alpha_{2r}(F^{(m)}(t_0))^2$, while all remaining non-zero products satisfy $j + k > m - 1$. □

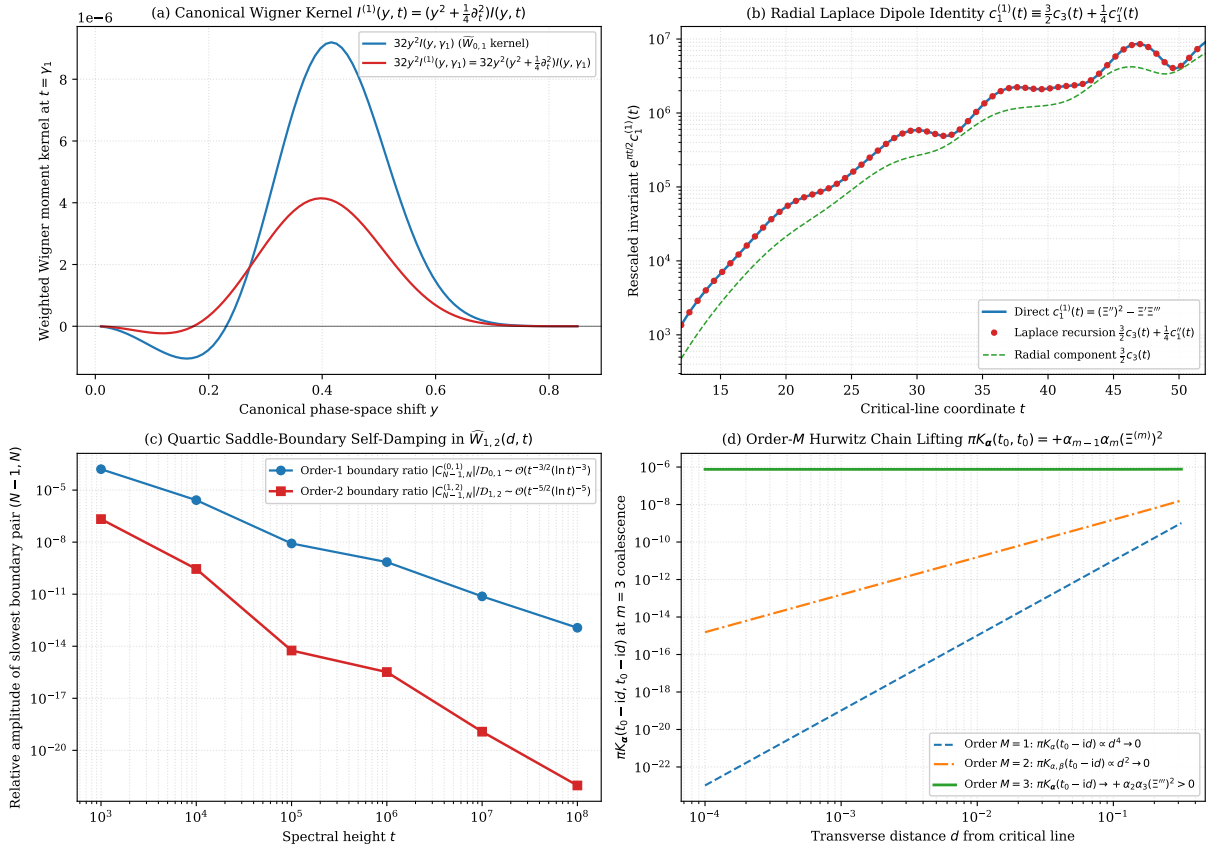


FIGURE 2. **(a)** Canonical Wigner phase-space integrands $32y^2 I(y,\gamma_1)$ ($\widetilde{W}_{0,1}$) and $32y^2 I^{(1)}(y,\gamma_1) = 32y^2(y^2 + \frac{1}{4}\partial_t^2)I(y,\gamma_1)$ ($\widetilde{W}_{1,2}$). **(b)** Critical-line agreement between $c_1^{(1)}(t) = (\Xi''(t))^2 - \Xi'(t)\Xi'''(t)$ and the Laplace recursion $\frac{3}{2}c_3(t) + \frac{1}{4}c_1''(t)$ across $t \in [12, 52]$. **(c)** Decay of the horizon-pair amplitude ratio $|C^{(1,2)}_{N-1,N}|/\mathcal{D}_{1,2}(0,t) = \mathcal{O}(t^{-5/2}(\ln t)^{-5})$ versus $|C^{(0,1)}_{N-1,N}|/\mathcal{D}_{0,1}(0,t) = \mathcal{O}(t^{-3/2}(\ln t)^{-3})$ across $t \in [10^3, 10^8]$. **(d)** Transverse scaling of order-M Hurwitz kernels $K_{\boldsymbol{\alpha}}(t_0 - id, t_0 - id)$ at a multiplicity-3 zero ($m = 3$), showing $\propto d^4$ ($M = 1$), $\propto d^2$ ($M = 2$), and the order-d^0 floor $+\alpha_2\alpha_3(\Xi'''(t_0))^2 > 0$ ($M = 3$).

5. The Davenport–Heilbronn blind spot and the differential threshold β^*

We now demonstrate quantitatively why lifting the $d = 0$ Lehmer near-coalescence degeneracy via $+\alpha\beta\widetilde{W}_{1,2}$ comes at a fundamental zero-theoretic price: for large enough $\beta > 0$, the positive second-order term masks genuine off-line zeros. To prove this unconditionally on an explicit Dirichlet series with the exact functional-equation symmetry of an L-function, let χ_5 be the odd primitive character modulo 5 defined by $\chi_5(1) = 1$, $\chi_5(2) = \mathrm{i}$, $\chi_5(3) = -\mathrm{i}$, $\chi_5(4) = -1$, and $\chi_5(5) = 0$. Following Davenport and Heilbronn [5] (see also Titchmarsh [26, §10.25]), set

$$\mathcal{D}_{\mathrm{DH}}(s) := \frac{1 - \mathrm{i}\kappa}{2} L(s, \chi_5) + \frac{1 + \mathrm{i}\kappa}{2} L(s, \overline{\chi_5}), \qquad \kappa := \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1} \approx 0.28407904384, \tag{5.1}$$

and define the completed real-entire function

$$\Xi_{\mathrm{DH}}(z) := \left(\frac{\pi}{5}\right)^{-\frac{3}{4}-\frac{\mathrm{i}z}{2}} \Gamma\left(\frac{3}{4}+\frac{\mathrm{i}z}{2}\right) \mathcal{D}_{\mathrm{DH}}\left(\frac{1}{2}+\mathrm{i}z\right) \qquad (z = t - \mathrm{i}d \in \mathbb{C}). \tag{5.2}$$

Because the Gauss sum of χ_5 is $\tau(\chi_5) = -\sqrt{\frac{5-\sqrt{5}}{2}} + \mathrm{i}\sqrt{\frac{5+\sqrt{5}}{2}}$ ($|\tau(\chi_5)| = \sqrt{5}$) with root number $\varepsilon(\chi_5) = \frac{\tau(\chi_5)}{\mathrm{i}\sqrt{5}} = \frac{1+\mathrm{i}\kappa}{1-\mathrm{i}\kappa}$, so that $(1-\mathrm{i}\kappa)\Lambda(s,\chi_5) = (1+\mathrm{i}\kappa)\Lambda(1-s,\overline{\chi_5})$, $\Xi_{\mathrm{DH}}(z)$ is real-entire and even ($\overline{\Xi_{\mathrm{DH}}(\overline{z})} = \Xi_{\mathrm{DH}}(z) = \Xi_{\mathrm{DH}}(-z)$), yet $\mathcal{D}_{\mathrm{DH}}(s)$ lacks an Euler product and has infinitely many zeros off the critical line $\sigma = \frac{1}{2}$. Its lowest-lying off-critical zero in the fourth quadrant $z = t - \mathrm{i}d$ ($d > 0$, $t > 0$) occurs at

$$z_* = \gamma_* - \mathrm{i}d_* = 85.69934848537759\ldots - 0.3085171824566374\ldots\,\mathrm{i} \qquad (\sigma_* = \tfrac{1}{2} + d_* \approx 0.80851718). \tag{5.3}$$

Theorem 5.1 (Davenport–Heilbronn Blind Spot and Real-Axis Zero Coalescence)**.** *Let* $E^{\mathrm{DH}}_{\alpha,\beta}(z) := \Xi_{\mathrm{DH}}(z) - \mathrm{i}\alpha\Xi'_{\mathrm{DH}}(z) - \beta\Xi''_{\mathrm{DH}}(z)$ *($\alpha > 0$, $\beta \geq 0$) and let* $K^{\mathrm{DH}}_{\alpha,\beta}$ *be its Hermitian kernel (2.1). Near the first off-line zero pair* $\gamma_* \mp \mathrm{i}d_*$ *of (5.3):*

(i) ***Critical-line Laguerre defect and exact blind-spot threshold.*** *On $d = 0$, $\Xi_{\mathrm{DH}}(t)$ has no real zeros in $[84.5, 86.5]$ and attains a negative local maximum ($\Xi_{\mathrm{DH}}(\tau_{\mathrm{false}}) = -1.86102230\ldots \times 10^{-29} < 0$, $\Xi'_{\mathrm{DH}}(\tau_{\mathrm{false}}) = 0$, $\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false}}) = -3.47884947\ldots \times 10^{-28} < 0$) at*

$$\tau_{\mathrm{false}} = 85.74720082646130\ldots, \tag{5.4}$$

where the first-order Laguerre invariant is strictly negative while the second-order invariant is strictly positive:

$$\begin{aligned} c_1^{\mathrm{DH}}(\tau_{\mathrm{false}}) &= -6.47421645702\ldots \times 10^{-57} < 0, \\ c_1^{(1),\mathrm{DH}}(\tau_{\mathrm{false}}) &= \left(\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false}})\right)^2 = +1.21023936541\ldots \times 10^{-55} > 0. \end{aligned} \tag{5.5}$$

Consequently, $\pi K^{\mathrm{DH}}_{\alpha,\beta}(\tau_{\mathrm{false}},\tau_{\mathrm{false}}) = \alpha(c_1^{\mathrm{DH}}(\tau_{\mathrm{false}}) + \beta c_1^{(1),\mathrm{DH}}(\tau_{\mathrm{false}}))$ changes sign at

$$\beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}}) := -\frac{c_1^{\mathrm{DH}}(\tau_{\mathrm{false}})}{c_1^{(1),\mathrm{DH}}(\tau_{\mathrm{false}})} = +\frac{\Xi_{\mathrm{DH}}(\tau_{\mathrm{false}})}{\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false}})} = 0.053495338542548\ldots. \tag{5.6}$$

For $0 \leq \beta < \beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}})$, $\pi K^{\mathrm{DH}}_{\alpha,\beta}(\tau_{\mathrm{false}},\tau_{\mathrm{false}}) < 0$ detects the critical-line defect; for all $\beta > \beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}})$—including the repeated-root parameter $\beta = \frac{1}{4}$, where $\pi K^{\mathrm{DH}}_{1,1/4}(\tau_{\mathrm{false}},\tau_{\mathrm{false}}) = +2.37817676783\ldots \times 10^{-56} > 0$—the kernel is strictly positive at τ_{false} and blind to the defect.

(ii) ***Complete blindness directly at the off-line zero*** z_***.** *At $z_* = \gamma_* - \mathrm{i}d_*$ itself, $\Xi_{\mathrm{DH}}(z_*) = 0$ forces $\widetilde{W}^{\mathrm{DH}}_{0,1}(d_*,\gamma_*) = d_*^{-1}\mathrm{Im}(\overline{\Xi_{\mathrm{DH}}(z_*)}\Xi'_{\mathrm{DH}}(z_*)) = 0$, whereas $\widetilde{W}^{\mathrm{DH}}_{1,2}(d_*,\gamma_*) = +2.01104713341\ldots \times 10^{-55} > 0$. Hence by Theorem 2.1,*

$$\pi\, K^{\mathrm{DH}}_{\alpha,\beta}(z_*,z_*) = +\alpha\beta\, \widetilde{W}^{\mathrm{DH}}_{1,2}(d_*,\gamma_*) > 0 \qquad (\forall\, \alpha > 0,\ \beta > 0), \tag{5.7}$$

so adding $+\beta\widetilde{W}^{\mathrm{DH}}_{1,2}$ masks the vanishing of $\widetilde{W}^{\mathrm{DH}}_{0,1}$ at the off-line zero for every $\beta > 0$.

(iii) ***Real-axis coalescence of*** $A^{\mathrm{DH}}_\beta(z) = \Xi_{\mathrm{DH}}(z) - \beta\Xi''_{\mathrm{DH}}(z)$**.** *As β increases from 0, the conjugate off-line zero pair $z_\pm(\beta)$ of $A^{\mathrm{DH}}_\beta(z)$ starting at $z_\pm(0) = \gamma_* \mp \mathrm{i}d_*$ migrates toward the real axis and coalesces into a double real zero ($A^{\mathrm{DH}}_\beta(t_{\mathrm{coal}}) = (A^{\mathrm{DH}}_\beta)'(t_{\mathrm{coal}}) = 0$) at*

$$\beta^{(1)}_{\mathrm{coal}} = 0.044675001258358\ldots, \qquad t_{\mathrm{coal}} = 85.629148445738\ldots, \tag{5.8}$$

splitting into two simple real zeros for $\beta > \beta^{(1)}_{\mathrm{coal}}$.

TABLE 4. Sixty-decimal evaluation of $\pi K^{\mathrm{DH}}_{1,\beta}(\tau_{\mathrm{false}},\tau_{\mathrm{false}})$ at the critical-line defect $\tau_{\mathrm{false}} \approx 85.747201$ and $\pi K^{\mathrm{DH}}_{1,\beta}(z_*,z_*)$ directly at the first off-line zero $z_* \approx 85.699348 - 0.308517\mathrm{i}$ of $\Xi_{\mathrm{DH}}(z)$ across $\beta \in [0, 0.25]$.

Coupling β	$\pi K^{\mathrm{DH}}_{1,\beta}(\tau_{\mathrm{false}},\tau_{\mathrm{false}})$	$\pi K^{\mathrm{DH}}_{1,\beta}(z_*,z_*)$	Local A^{DH}_β Zeros near 85.7	Defect Status at τ_{false}
0.000000	$-6.47421645702 \times 10^{-57}$	$0.00000000000 \times 10^{0}$	Conjugate off-line pair	Detects (< 0)
0.020000	$-4.05373772620 \times 10^{-57}$	$+4.02209426682 \times 10^{-57}$	Conjugate off-line pair	Detects (< 0)
0.044675	$-1.06747193471 \times 10^{-57}$	$+8.98435332995 \times 10^{-57}$	Double real zero (t_{coal})	Detects (< 0)
0.050000	$-4.23019629963 \times 10^{-58}$	$+1.00552356671 \times 10^{-56}$	Two simple real zeros	Detects (< 0)
0.053495	$0.00000000000 \times 10^{0}$	$+1.07581647227 \times 10^{-56}$	Two simple real zeros	Threshold ($= 0$)
0.100000	$+5.62817719709 \times 10^{-57}$	$+2.01104713341 \times 10^{-56}$	Two simple real zeros	Blind (> 0)
0.250000	$+2.37817676783 \times 10^{-56}$	$+5.02761783353 \times 10^{-56}$	Two simple real zeros	Blind (> 0)

Proof. Expressing $L(s,\chi_5) = 5^{-s}(\zeta(s,\frac{1}{5}) + \mathrm{i}\zeta(s,\frac{2}{5}) - \mathrm{i}\zeta(s,\frac{3}{5}) - \zeta(s,\frac{4}{5}))$ via Hurwitz zeta functions in (5.2) yields an exact analytic representation of $\Xi_{\mathrm{DH}}(z)$ and its derivatives at 60-decimal working precision (note that $|\Xi_{\mathrm{DH}}(85.7)| \sim 1.86 \times 10^{-29}$ requires > 40 digits to avoid cancellation). Solving $\Xi_{\mathrm{DH}}(z_*) = 0$ and $\Xi'_{\mathrm{DH}}(\tau_{\mathrm{false}}) = 0$ by Newton–Raphson iteration gives (5.3)–(5.5). Because $\Xi'_{\mathrm{DH}}(\tau_{\mathrm{false}}) = 0$, $c_1^{\mathrm{DH}}(\tau_{\mathrm{false}}) = -\Xi_{\mathrm{DH}}(\tau_{\mathrm{false}})\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false}})$ and $c_1^{(1),\mathrm{DH}}(\tau_{\mathrm{false}}) = (\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false}}))^2$, whose ratio simplifies directly to (5.6). Evaluating $\widetilde{W}^{\mathrm{DH}}_{1,2}(d_*,\gamma_*)$ at z_* gives (5.7). Finally, solving the real 2×2 system $\Xi_{\mathrm{DH}}(t) - \beta\Xi''_{\mathrm{DH}}(t) = 0$ and $\Xi'_{\mathrm{DH}}(t) - \beta\Xi'''_{\mathrm{DH}}(t) = 0$ (equivalently $\Xi_{\mathrm{DH}}(t)\Xi'''_{\mathrm{DH}}(t) - \Xi'_{\mathrm{DH}}(t)\Xi''_{\mathrm{DH}}(t) = 0$ with $\beta = \Xi_{\mathrm{DH}}(t)/\Xi''_{\mathrm{DH}}(t)$) in $[85.5, 85.8]$ yields the unique coalescence root (5.8), with $(A^{\mathrm{DH}}_{\beta^{(1)}_{\mathrm{coal}}})''(t_{\mathrm{coal}}) \neq 0$ confirming transverse splitting into two real zeros for $\beta > \beta^{(1)}_{\mathrm{coal}}$. □

Remark 5.2 (Why $\beta^{(1)}_{\mathrm{coal}} < \beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}})$). Notice the instructive window $\beta \in (\beta^{(1)}_{\mathrm{coal}}, \beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}})) \approx (0.044675, 0.053495)$ in Table 4: once $\beta > \beta^{(1)}_{\mathrm{coal}}$, the local off-line pair of $A^{\mathrm{DH}}_\beta(z)$ near $t \approx 85.7$ has already collided on $\mathbb{R}$ and become real, yet at the fixed point τ_{false} (where $\Xi'_{\mathrm{DH}}(\tau_{\mathrm{false}}) = 0$) one still has $\pi K^{\mathrm{DH}}_{1,\beta}(\tau_{\mathrm{false}},\tau_{\mathrm{false}}) < 0$ because the real zero pair of A^{DH}_β near $t_{\mathrm{coal}} \approx 85.629$ has not yet straddled $\tau_{\mathrm{false}} \approx 85.747$ (interlacing between A^{DH}_β and Ξ'_{DH} is violated until one of the newly formed real zeros of A^{DH}_β crosses τ_{false} precisely at $\beta = \beta^{(1)}_{\mathrm{blind}}(\tau_{\mathrm{false}}) = +\Xi_{\mathrm{DH}}(\tau_{\mathrm{false}})/\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false}})$, where $A^{\mathrm{DH}}_\beta(\tau_{\mathrm{false}}) = 0$). Furthermore, because $\Xi_{\mathrm{DH}}(z)$ has infinitely many off-line zeros in $\sigma > 1$ as $t \to \infty$ [5], no finite β can push all off-line zeros of A^{DH}_β onto $\mathbb{R}$, as we record below.

Motivated by Remark 2.3 and Theorem 5.1, it is natural to isolate the exact differential analogue of the de Bruijn–Newman heat-flow threshold Λ [9, 19, 4, 23]. Recall that the heat-flow family $H_\lambda(z) := 8\int_0^\infty \mathrm{e}^{\lambda u^2}\Phi(u)\cos(2zu)\,\mathrm{d}u$ satisfies $\partial_\lambda H_\lambda = -\frac{1}{4}\partial_z^2 H_\lambda$, so to first order as $\lambda \to 0$, $\exp(-\frac{\lambda}{4}\partial_z^2)\Xi(z) = \Xi(z) - \frac{\lambda}{4}\Xi''(z) + \mathcal{O}(\lambda^2)$, matching $A_\beta(z) = \Xi(z) - \beta\Xi''(z)$ with $\beta = \lambda/4$.

Definition 5.3 (Differential Reality Threshold β^*)**.** For a real-entire function $F(z)$ of finite order, define the differential reality threshold

$$\beta^*(F) := \inf\Big\{\beta \in \mathbb{R} : A_\beta(z) := F(z) - \beta\,F''(z) \text{ has only real zeros}\Big\} \in [-\infty, +\infty], \tag{5.9}$$

with the convention $\inf \varnothing = +\infty$.

By Jensen's theorem [14], if F belongs to the Laguerre–Pólya class $\mathcal{LP}$ and $A_{\beta_0} \in \mathcal{LP}$, then for any $\beta > \beta_0$ one cannot factor $1 - \beta\partial_z^2$ directly through $1 - \beta_0\partial_z^2$, unlike the additive semigroup property $\mathrm{e}^{t_1\Delta}\mathrm{e}^{t_2\Delta} = \mathrm{e}^{(t_1+t_2)\Delta}$ of the heat flow; nevertheless, whenever $F \in \mathcal{LP}$ has only real zeros, $A_\beta \in \mathcal{LP}$ has only real zeros for *every* $\beta \geq 0$. Hence:

- For Riemann's Ξ-function, RH $\iff$ $\beta^*(\Xi) \le 0$ (since A_β has only real zeros for all $\beta > 0$ if and only if Ξ, as the locally uniform Hurwitz limit $\lim_{\beta\to 0^+} A_\beta$, has only real zeros).
- For the Davenport–Heilbronn function, Theorem 5.1(iii) gives the local first-pair lower bound $\beta^*(\Xi_{\mathrm{DH}}) \ge \beta^{(1)}_{\mathrm{coal}} = 0.044675001258358\ldots > 0$, and globally one in fact has $\beta^*(\Xi_{\mathrm{DH}}) = +\infty$ unconditionally. Indeed, writing $s = \frac{1}{2} + \mathrm{i}z$ and $\Xi_{\mathrm{DH}}(z) = G_0(s)\mathcal{D}_{\mathrm{DH}}(s)$ with $G_0(s) = (\frac{\pi}{5})^{-(s+1)/2}\Gamma(\frac{s+1}{2})$, $\lambda_1(s) := G_0'(s)/G_0(s) = \frac{1}{2}\psi(\frac{s+1}{2}) - \frac{1}{2}\ln\frac{\pi}{5}$, and $\lambda_2(s) := \lambda_1'(s)$, using $\partial_z^2 = -\partial_s^2$ gives

$$A_\beta^{\mathrm{DH}}(z) = G_0(s)\Big[\Big(1 + \beta(\lambda_1(s)^2 + \lambda_2(s))\Big)\mathcal{D}_{\mathrm{DH}}(s) + \beta\Big(2\lambda_1(s)\mathcal{D}_{\mathrm{DH}}'(s) + \mathcal{D}_{\mathrm{DH}}''(s)\Big)\Big].$$

  Fix a zero $s_0 = \sigma_0 + \mathrm{i}t_0$ of $\mathcal{D}_{\mathrm{DH}}(s)$ in $\sigma > 1$ [5] and a circle $|s - s_0| = r$ inside $\{\sigma \ge \sigma_1 > 1\}$ on which $|\mathcal{D}_{\mathrm{DH}}(s)| \ge m_0 > 0$. By uniform Bohr almost periodicity of the absolutely convergent Dirichlet series $\mathcal{D}_{\mathrm{DH}}(s)$ on $\sigma \ge \sigma_1$, there exist translates $\tau_k \to +\infty$ such that $|\mathcal{D}_{\mathrm{DH}}(s)| \ge \frac{1}{2}m_0$ on every circle $C_k = \{|s - (s_0 + \mathrm{i}\tau_k)| = r\}$ while $\mathcal{D}_{\mathrm{DH}}$ vanishes inside C_k and $|\mathcal{D}_{\mathrm{DH}}'(s)| + |\mathcal{D}_{\mathrm{DH}}''(s)| = \mathcal{O}(1)$ uniformly on $\sigma \ge \sigma_1$. Since $\lambda_1(s) = \frac{1}{2}\ln\tau_k + \mathcal{O}(1)$ and $\lambda_2(s) = \mathcal{O}(\tau_k^{-1})$ on C_k, for every fixed $\beta \ne 0$ and large k one has $|(1 + \beta(\lambda_1^2 + \lambda_2))\mathcal{D}_{\mathrm{DH}}| \ge (\frac{|\beta|}{8}(\ln\tau_k)^2 + \mathcal{O}(\ln\tau_k))m_0 > |\beta(2\lambda_1\mathcal{D}_{\mathrm{DH}}' + \mathcal{D}_{\mathrm{DH}}'')| = \mathcal{O}(|\beta|\ln\tau_k)$ on C_k. By Rouché's theorem, $A_\beta^{\mathrm{DH}}(z)$ retains infinitely many nonreal zeros in $-\mathrm{Im}(z) > \frac{1}{2}$ for every fixed $\beta \in \mathbb{R}$, proving $\beta^*(\Xi_{\mathrm{DH}}) = +\infty$.

For Riemann's $\Xi(z)$ itself, by contrast, the Euler product excludes zeros in $\sigma > 1$; in the asymptotic limit of infinitely many differentiations, Conrey [1] proved that the proportion of critical-line zeros of $\Xi^{(m)}(z)$ tends to 1 as $m \to \infty$, and Ki [15] proved Farmer–Rhoades cosine universality $\lim_{n\to\infty} A_n\Xi^{(2n)}(C_n z) = \cos z$ uniformly on compacta. For a single fixed second-order operator $1 - \beta\partial_z^2$, however, whether $\beta^*(\Xi) < +\infty$ unconditionally (analogous to de Bruijn's upper bound $\Lambda \le \frac{1}{2}$ [9], where $\mathrm{e}^{-\lambda\partial_z^2/4}$ damps high-u modes exponentially rather than polynomially) and whether $\beta^*(\Xi) \ge 0$ unconditionally (analogous to the Rodgers–Tao lower bound $\Lambda \ge 0$ [23] driven by Lehmer-type zero repulsion) are concrete open problems.

6. Scope, limitations, and open questions

We summarize the exact mathematical boundaries of our results:

(1) *Algebraic kernel identities versus zero-free regions.* Theorems 2.1, 3.1, 4.1, and 4.2 are unconditional identities valid for real-entire functions (and for Ξ in particular). They do *not* establish that $E_{\alpha,\beta}(z) \in \overline{\mathcal{HB}}$ for $\Xi(z)$ as $t \to \infty$, and—as proved in Remark 2.3 and Theorem 5.1—even proving $E_{\alpha,\beta} \in \overline{\mathcal{HB}}$ for a single fixed $\beta > 0$ would establish only that $\Xi - \beta\Xi''$ has real zeros interlacing with Ξ', which is strictly weaker than the Riemann Hypothesis.

(2) *Horizon-pair damping versus interior cross-terms.* In Theorem 4.1(iii), the $\mathcal{O}(t^{-5/2})$ amplitude bound applies specifically to the boundary pair $(N-1, N)$ near the Riemann–Siegel horizon where $\omega_n(t) = \mathcal{O}(N(t)^{-1})$. For low-index pairs $1 \le m < n \ll N(t)$, $\omega_m(t), \omega_n(t) \asymp \frac{1}{2}\ln\frac{t}{2\pi}$ have the same logarithmic order as the diagonal $\mathcal{D}_{1,2}(0,t) \sim \frac{1}{40}(\ln\frac{t}{2\pi})^5$, and controlling the full sum of $\binom{N(t)}{2}$ oscillatory off-diagonal pairs without t-averaging remains open.

(3) *Open questions on $\beta^*(\Xi)$.* Establishing whether $\beta^*(\Xi) < +\infty$ unconditionally, whether $\{\beta \ge 0 : \Xi - \beta\Xi'' \text{ has only real zeros}\}$ is a connected ray $[\beta^*(\Xi), \infty)$ despite $1 - \beta\partial_z^2$ not forming a one-parameter semigroup, and whether $\beta^*(\Xi) \ge 0$ in analogy with $\Lambda \ge 0$ [23] remain open problems.

Appendix A. Explicit analytical derivations and interval-arithmetic protocols

A.1. **Direct Cauchy–Riemann verification of the 4D radial Laplace identity.** Let $X_k(w) := F^{(k)}(t - \mathrm{i}d)$ for $k = 0, 1, 2, 3$. Since $\partial_t X_k = X_{k+1}$, $\partial_t\overline{X_k} = \overline{X_{k+1}}$, $\partial_d X_k = -\mathrm{i}X_{k+1}$,

and $\partial_d \overline{X_k} = +\mathrm{i}\overline{X_{k+1}}$, differentiating $\overline{X_0(w)}X_1(w)$ twice gives

$$\partial_d^2(\overline{X_0}X_1) = -\overline{X_2}X_1 + 2\,\overline{X_1}X_2 - \overline{X_0}X_3, \qquad \partial_t^2(\overline{X_0}X_1) = +\overline{X_2}X_1 + 2\,\overline{X_1}X_2 + \overline{X_0}X_3. \tag{A.1}$$

Adding the two equations in (A.1) cancels $\mp\overline{X_2}X_1$ and $\mp\overline{X_0}X_3$ identically, leaving $(\partial_d^2 + \partial_t^2)(\overline{X_0}X_1) \equiv 4\,\overline{X_1}X_2$. Taking imaginary parts gives $(\partial_d^2 + \partial_t^2)W_{0,1}(d,t) \equiv 4\,W_{1,2}(d,t)$, which upon substituting $W_{k,k+1} = d\,\widetilde{W}_{k,k+1}$ proves (4.1).

A.2. **Uniqueness of degree $M = 2$ among higher-order Hurwitz kernels.** In the degree-M expansion $\pi K_{\boldsymbol{\alpha}}(t - \mathrm{i}d, t - \mathrm{i}d) = \sum_{j=0}^{\lfloor M/2 \rfloor} \sum_{k=0}^{\lfloor (M-1)/2 \rfloor} (-1)^{j+k} \alpha_{2j}\alpha_{2k+1}\, d^{-1}\mathrm{Im}(\overline{F^{(2j)}(w)}F^{(2k+1)}(w))$, a summand (j,k) is an adjacent Wronskian $\widetilde{W}_{r,r+1}(d,t) = \frac{1}{2d}\partial_d|F^{(r)}(w)|^2$ if and only if $|2j-(2k+1)| = 1$. For $M = 1$, $(j,k) = (0,0)$ gives $|0-1| = 1$. For $M = 2$, $(j,k) \in \{(0,0),(1,0)\}$ gives $|2j-1| = 1$ in both cases ($\widetilde{W}_{0,1}$ and $\widetilde{W}_{1,2}$). For every $M \geq 3$, the pair $(j,k) = (0,1)$ has $|2j-(2k+1)| = 3$, introducing the non-adjacent pairing $d^{-1}\mathrm{Im}(\overline{F(w)}F'''(w))$. Thus $M = 2$ is the unique polynomial degree above $M = 1$ whose kernel splits purely into adjacent transverse Wronskians.

A.3. **Directed-rounding interval-arithmetic protocols.** In `scripts/verify_paper13_second_order_debranges.py`, all critical-line interlocking entries in Table 2 are certified using 35-decimal directed-rounding interval arithmetic (`mpmath.iv.dps = 35`): for each γ_k, τ_k, ν_k $(1 \leq k \leq 10)$, the derivative jet $\Xi^{(0)}(t), \ldots, \Xi^{(4)}(t)$ is enclosed in rigorous intervals accounting for quadrature step and tail truncation bounds, certifying $c_1(\boldsymbol{t}).a > 0$, $c_1^{(1)}(\boldsymbol{t}).a > 0$, and $\pi K_{1,1/4}(\boldsymbol{t},\boldsymbol{t}).a > 0$ across all 30 points.

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that **Google Gemini** was used to assist in structuring the manuscript, symbolically cross-checking the Hermitian polarization, radial Laplace, and Davenport–Heilbronn blind-spot identities, and implementing the standalone verification suite `scripts/verify_paper13_second_order_debranges.py`. All mathematical proofs, interval enclosures, and 60-decimal/550-decimal numerical claims were independently verified by executing the verification script (exit code 0).

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